

Define Problem Statements Template

Date	15 July 2026
Team ID	XXXX
Project Name	AI Learning Assistant - Emotion-Aware Tutoring System
Maximum Marks	2 Marks

Define the problem statement(s) your project addresses, using the format: "I am (user) who is trying to (job) but (barrier) because (root cause) which makes me feel (emotion)."

PS-1	
I am (User)	an instructor, teaching assistant, or academic support team member
Who is trying to	interpret a student's emotional state from their written messages so I can respond with the right tone and guidance
But	manually reading messages, guessing emotions, and crafting an appropriately empathetic reply slows down every response cycle and can cause struggling students to be missed entirely
Because	there is no automated way to classify a student's emotional state (Confused, Frustrated, Bored, Curious, Confident) from plain text before I respond
Which makes me feel	unable to triage efficiently and worried that some students who need help the most are the ones I notice last

PS-2	
I am (User)	a student studying independently or self-paced through an online platform
Who is trying to	get quick, empathetic guidance when I'm stuck or unsure about a concept
But	manually searching forums or drafting a help request takes time and rarely captures how I'm actually feeling about the problem
Because	the platform has no mechanism to detect emotional cues like confusion or frustration directly from what I type, so every response feels generic rather than personalized
Which makes me feel	unsupported in the moment, and more likely to disengage quietly instead of asking for help

Summary

Both problem statements point to the same underlying gap: free-text learning interactions carry emotional signal (confusion, frustration, boredom, curiosity, confidence) that is never captured or acted on, whether the reader is a human instructor manually triaging messages or an automated system generating a response. The **AI-Driven Emotion Detection & Personalized Learning Support Platform** addresses this by combining deep learning emotion classifiers (BiLSTM and BERT) with rule-based keyword enhancement and Gemini-generated guidance, turning a learner's free-text description of their challenge directly into an emotion-aware, supportive response.